St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 07

Imaging of the Upper Limb

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Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1144	Explain why the glenohumeral joint can be easily dislocated.
SOM.MKII.BPM1.1.MSK.1.ANAT.1145	Describe shoulder dislocation injuries, particularly antero - inferior dislocations, and explain the nerve injuries that might accompany shoulder dislocations.
SOM.MKII.BPM1.1.MSK.1.ANAT.1146	Identify antero-inferior shoulder dislocation on a radiograph.
SOM.MKII.BPM1.1.MSK.1.ANAT.1147	Describe the anatomical and clinical signs of rotator cuff disorders due to injury of the supraspinatus muscle
SOM.MKII.BPM1.1.MSK.1.ANAT.1154	Identify the bones of the arm in imaging studies including and use this information to interpret clinical signs and symptoms.
SOM.MKII.BPM1.1.MSK.1.ANAT.1155	Identify the shoulder joints in imaging studies including but not limited to CT, MRI and radiographic images.
SOM.MKII.BPM1.1.MSK.1.ANAT.1162	Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.
SOM.MKII.BPM1.1.MSK.1.ANAT.1171	Identify the bones of the forearm in imaging studies including and use this information to interpret clinical signs and symptoms.
SOM.MKII.BPM1.1.MSK.1.ANAT.1172	Identify the elbow joint in imaging studies and use this information to interpret clinical signs and symptoms.
SOM.MKII.BPM1.1.MSK.1.ANAT.1173	Recognize and describe Colle's fracture and Smith's fracture on a radiograph.
SOM.MKII.BPM1.1.MSK.1.ANAT.1181	Describe and identify Boxer's fractures and possible clinical consequences.
SOM.MKII.BPM1.1.MSK.1.ANAT.1182	Identify the following carpal bone fractures: scaphoid fracture and fracture of the hook of the hamate, and describe their possible clinical consequences
SOM.MKII.BPM1.1.MSK.1.ANAT.1195	Identify the bones of the hand in imaging studies including but not limited to CT, MRI and radiographic images.
SOM.MKII.BPM1.1.MSK.1.ANAT.1196	Identify the shoulder wrist and hand joints in imaging studies including but not limited to CT, MRI and radiographic images.
SOM.MKII.BPM1.1.MSK.1.ANAT.1209	Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph.

Initial Imaging for Musculoskeletal Diagnostics

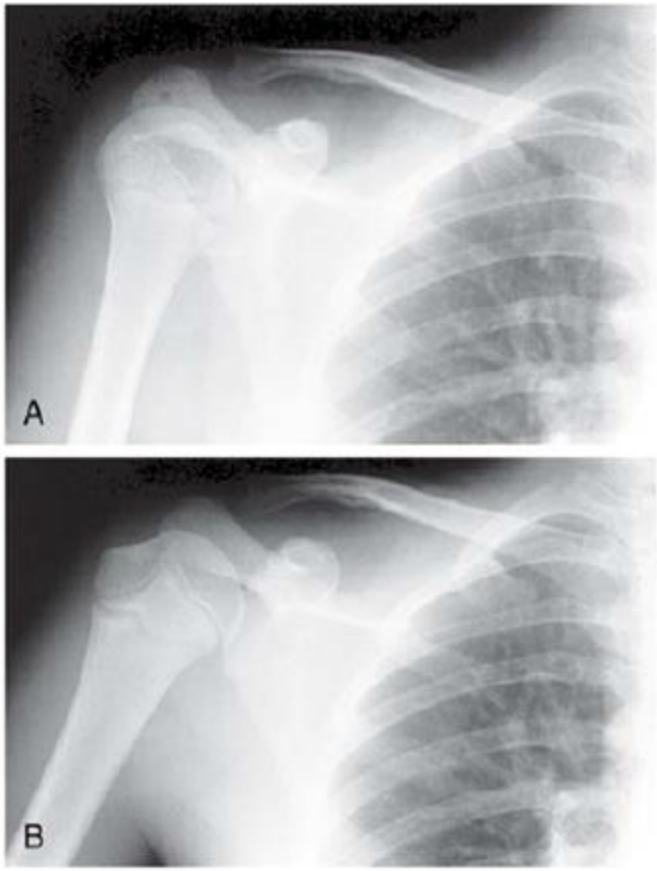
Condition	X-ray	CT	MRI
Joint pain	+		
Fracture	+		
Stress fracture	+		N
Spinal trauma	+	P	P
Osteomyelitis	+		N
Soft tissue mass	+		N
Elbow pain (chronic)	+		N
Shoulder pain - acute	+		N
Shoulder pain - persistent or postoperative			+

Table abbreviations:

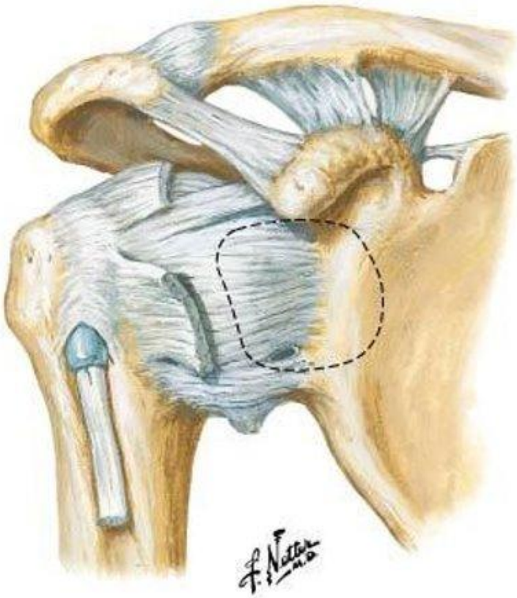
- +: initial option
- N: only if negative
- P: for possible consideration

Shoulder

Shoulder on Conventional Radiograph



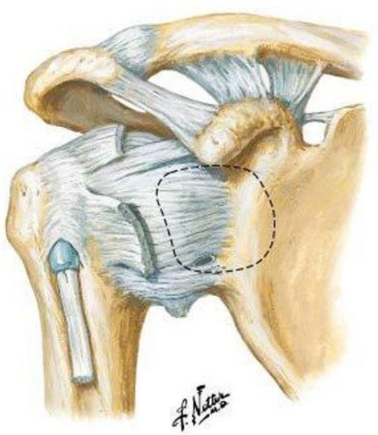
• Fig. 8.40

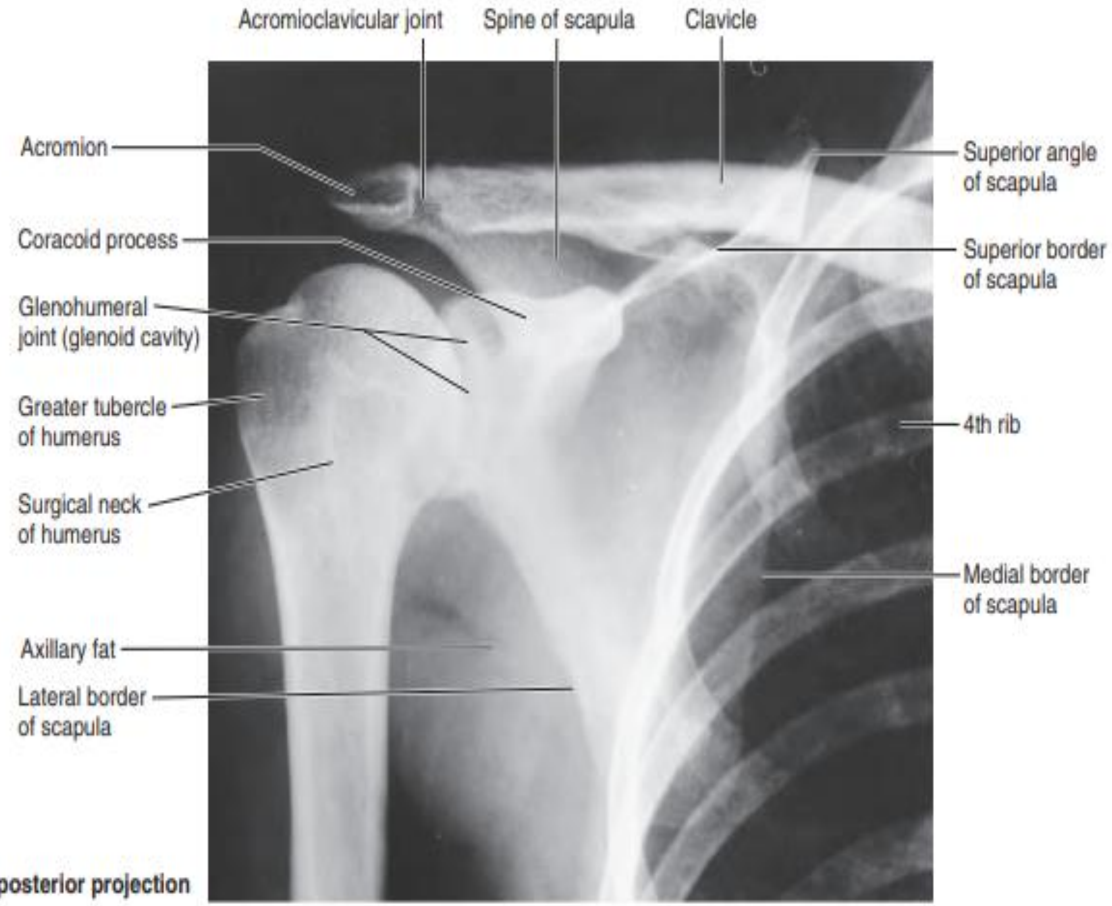


Conventional Radiograph Interpretation

Some Musculoskeletal X-ray Principles:

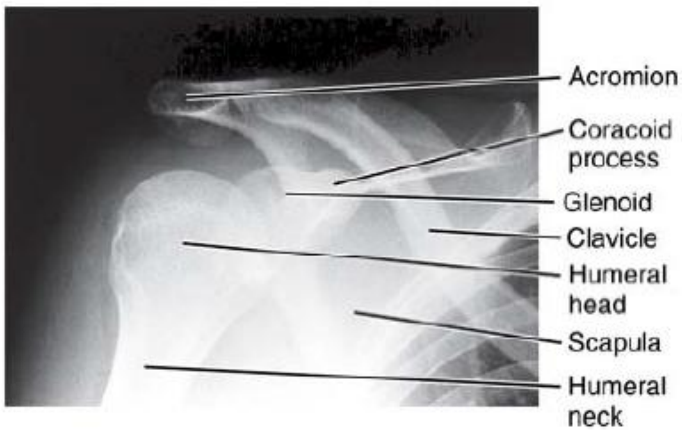
- **A**lignment
- **B**one
- **C**artilage
- **D**ensities/deformities
- **E**verything else



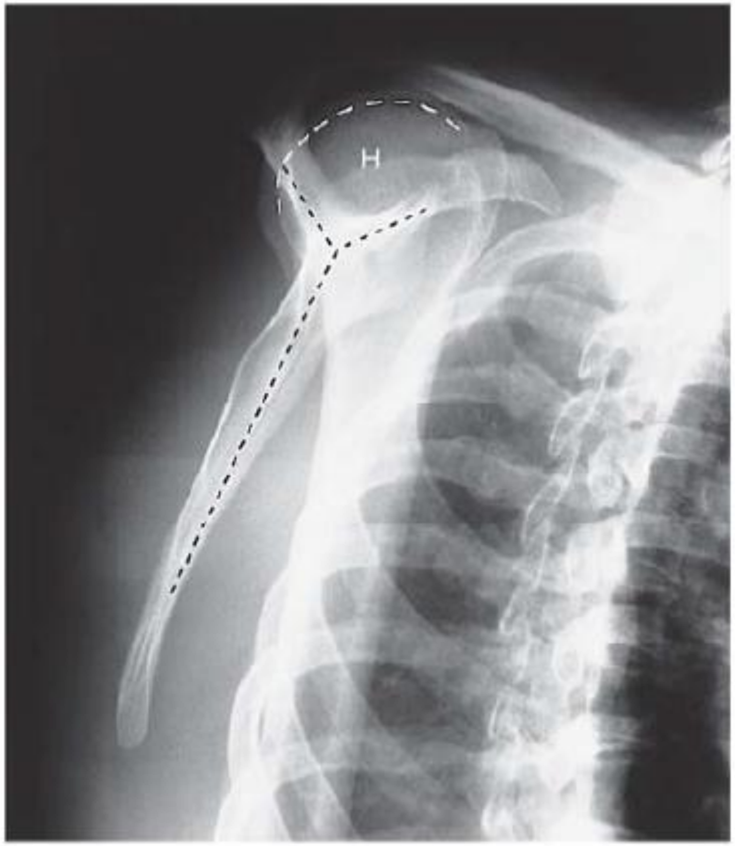


(B) Anteroposterior projection

Radiograph Positioning

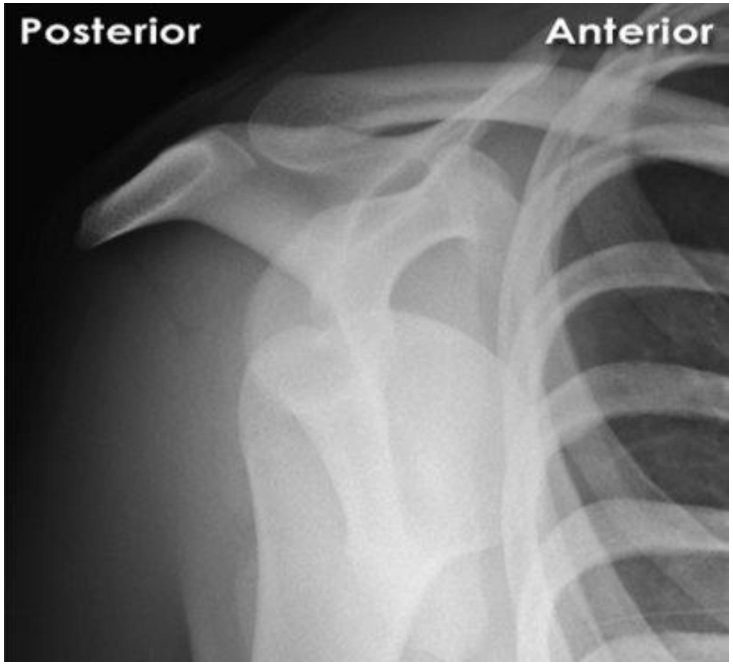


• Fig. 8.41



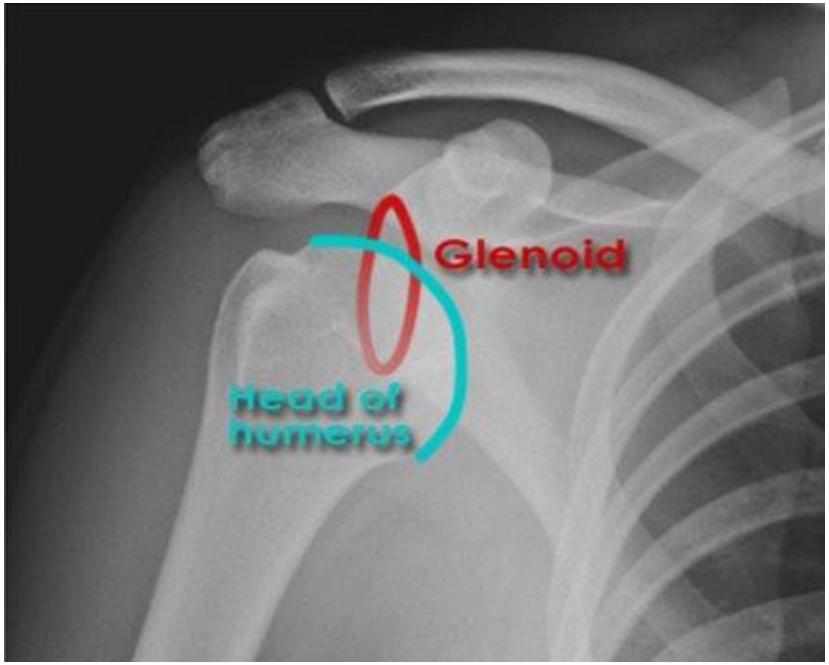
• Fig. 8.43

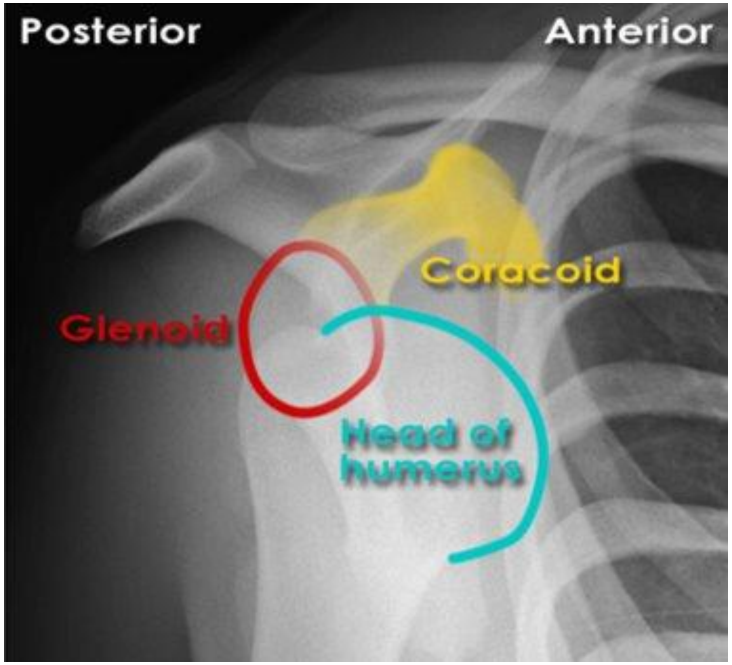
Clinical Correlate: Shoulder Dislocation



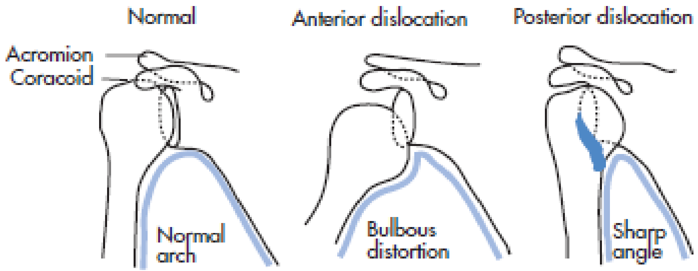
- What direction is the dislocation?
- What neurovascular structures may be damaged in this patient?

Clinical Correlate: Shoulder Dislocation



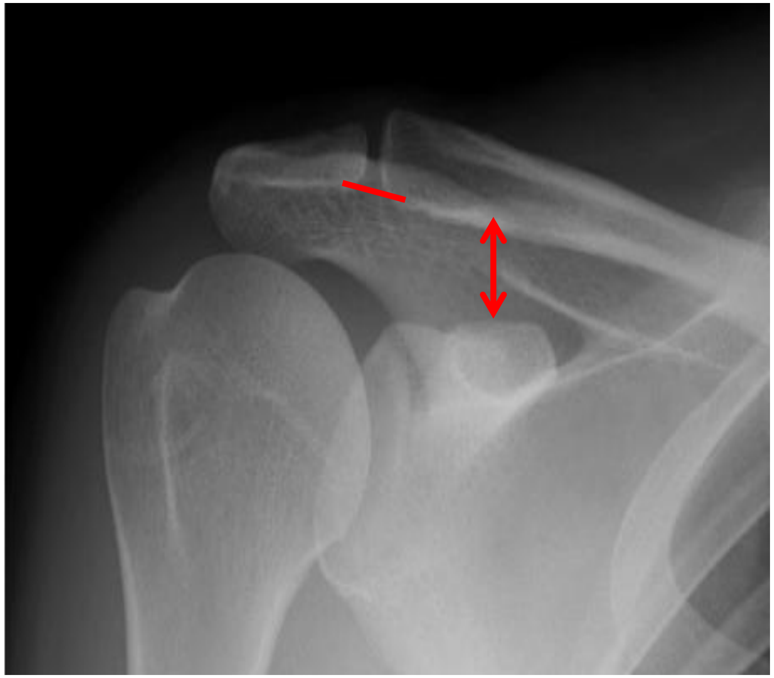


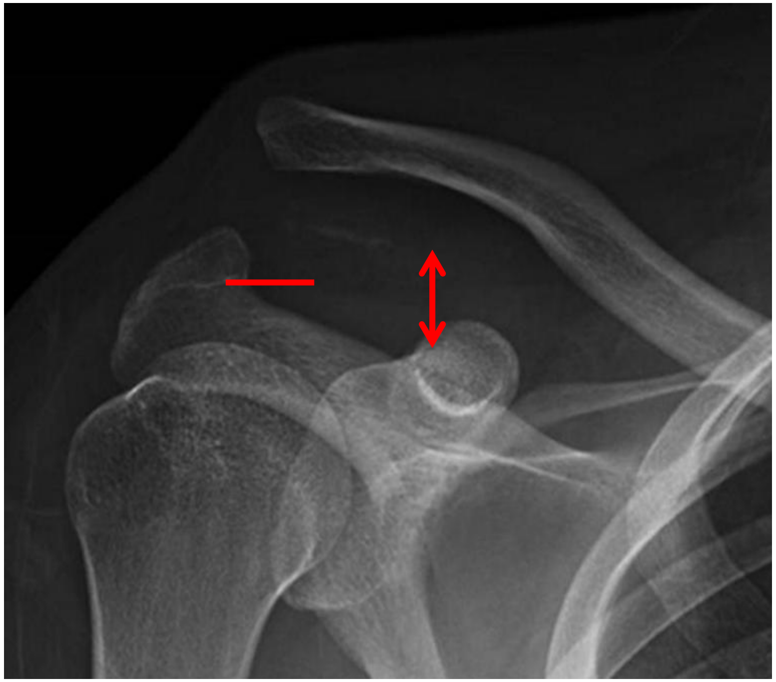
- What direction is the dislocation?
  - ***Anteroinferior***



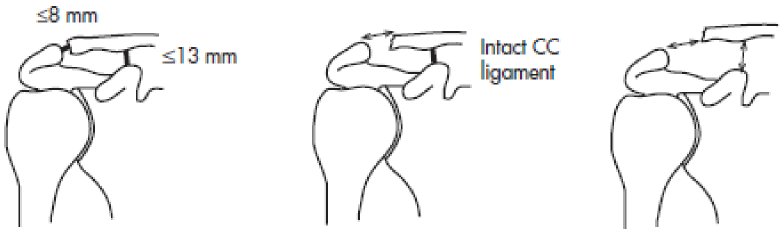
- What neurovascular structures may be damaged in this patient?
  - Axillary nerve
  - Posterior circumflex humeral artery

Clinical Correlate: Shoulder Separation





- Varying degrees of severity
- Tearing of the acromioclavicular joint/ligament **with or without** the coracoclavicular ligament
- Restricted range of shoulder movement



Acromioclavicular joint assessment by ultrasonography

TABLE 21-2 ADVANTAGES AND DISADVANTAGES OF ULTRASONOGRAPHY

Advantages	Disadvantages
No ionizing radiation	Difficulty penetrating through bone
No known long-term side effects	Gas-filled structures reduce its utility
"Real-time" images	Penetration may be difficult in obese patients
Produces little or no patient discomfort	Dependent on the skills of the operator doing the scanning
Small, portable, inexpensive, ubiquitous	

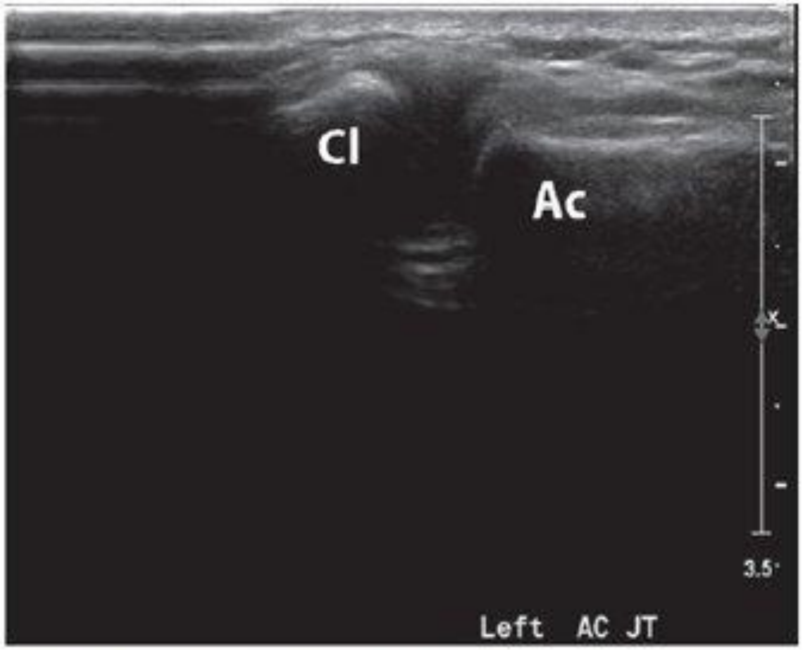


FIGURE 28.5 • Normal acromioclavicular joint. Superior shoulder sonogram shows the joint between the distal clavicle (Cl) and acromion (Ac).

Shoulder Joints: MRI

(C)

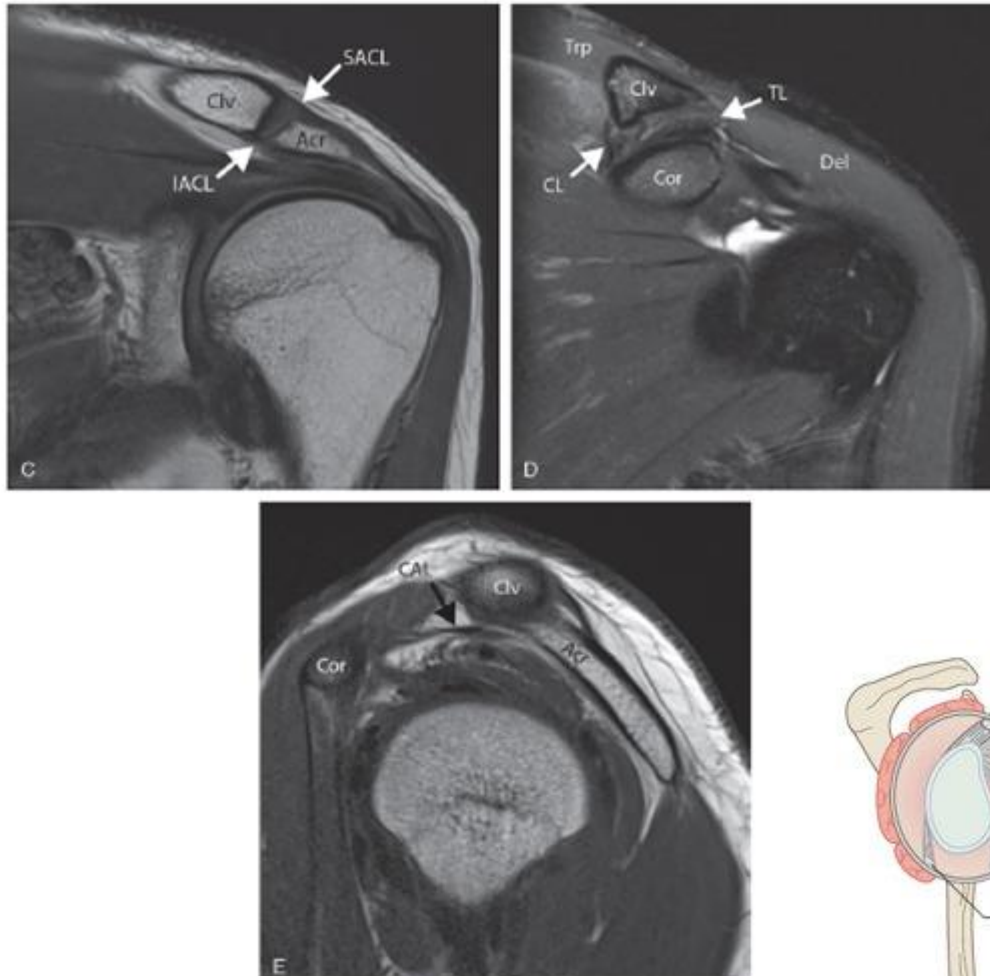


FIGURE 6.10

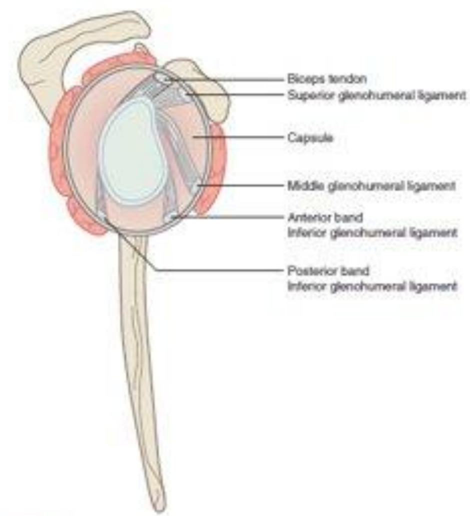
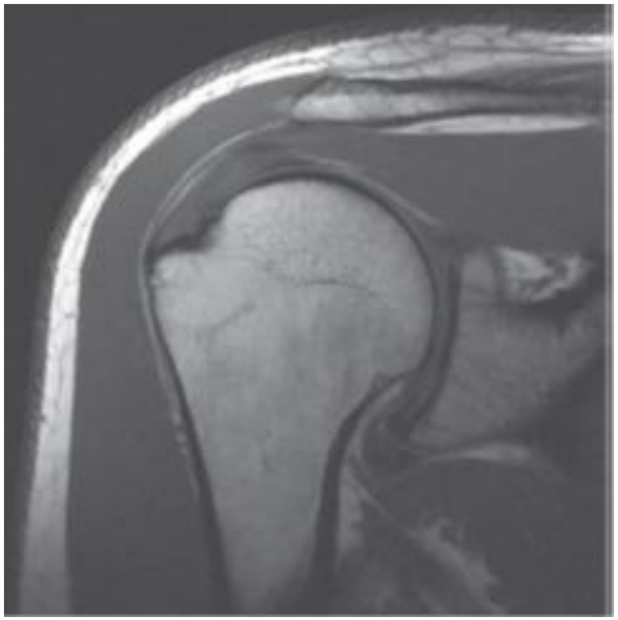
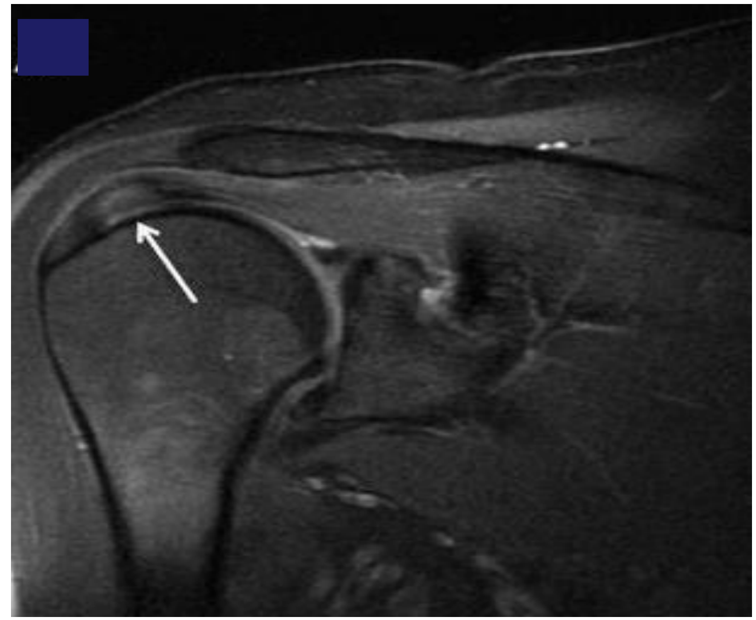


Fig. 15.15 Glenohumeral ligaments.

Clinical Correlate: Rotator Cuff Injuries



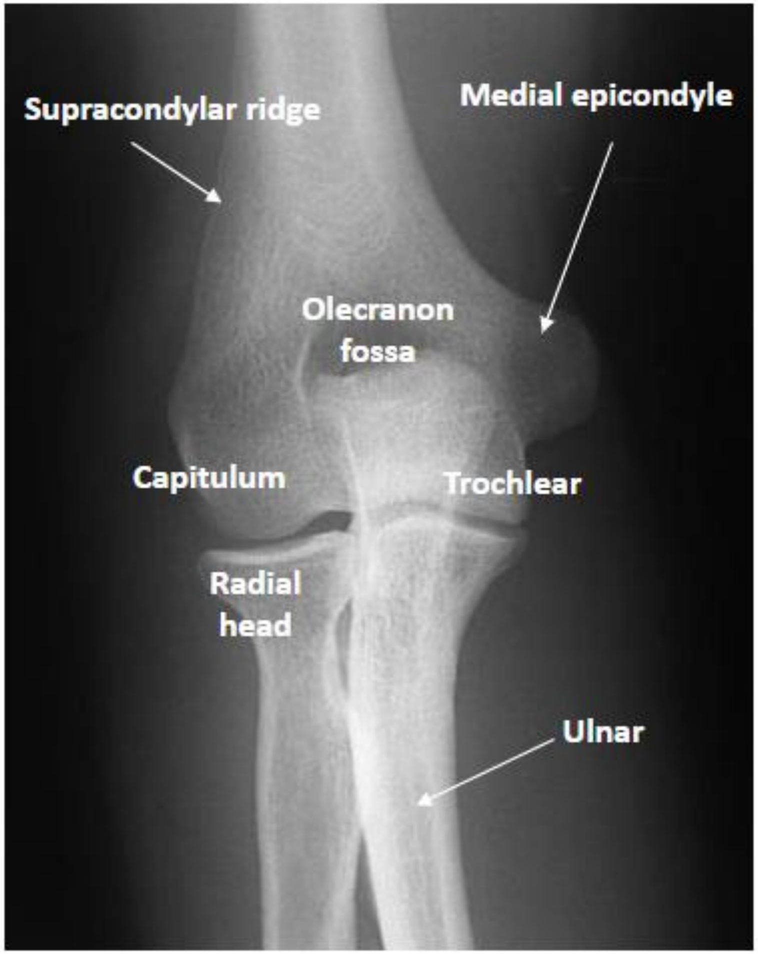


- Most commonly involves which muscle?
- Can be impinged by which ligament?
- What overlying structure can become inflamed?

Elbow

Elbow: Conventional Radiograph

Elbow – review



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Elbow Alignment

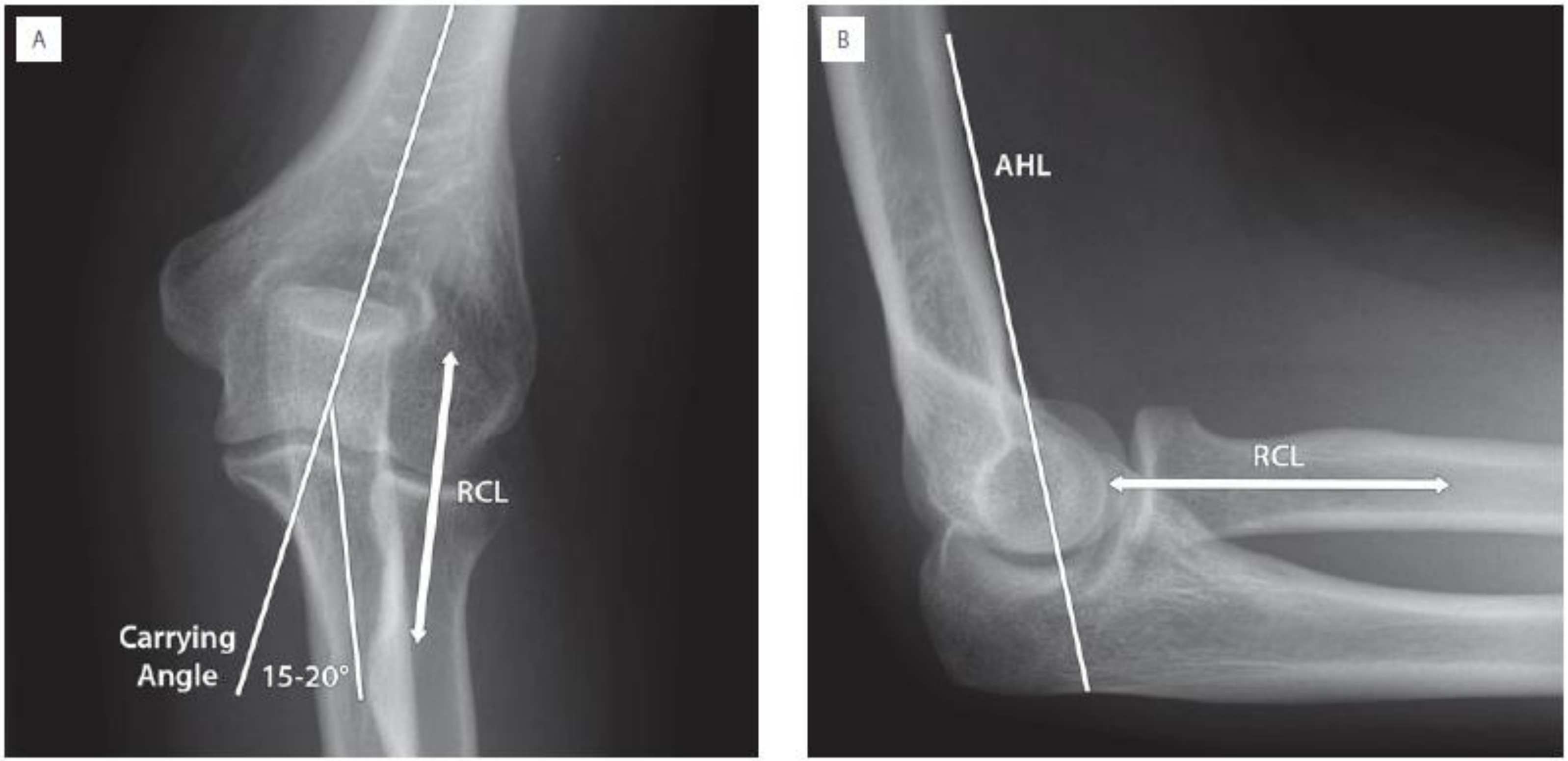
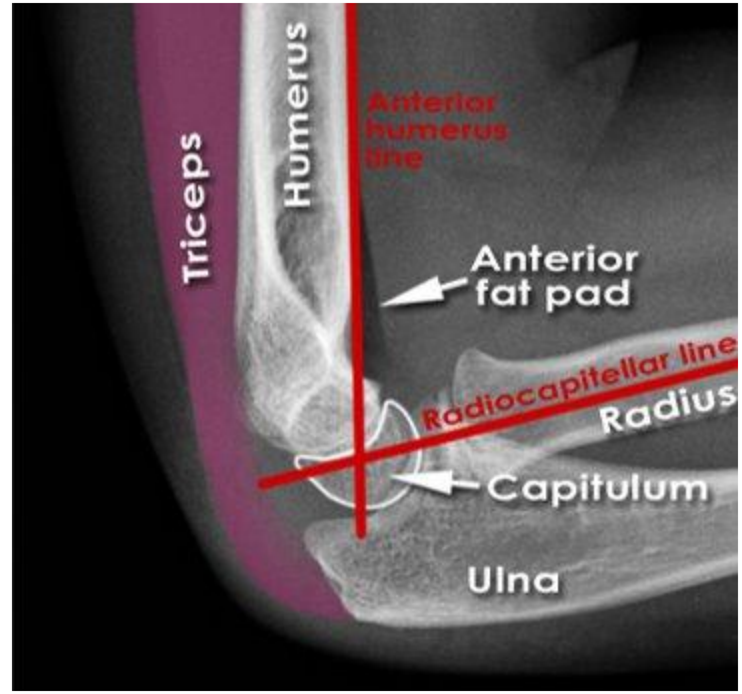


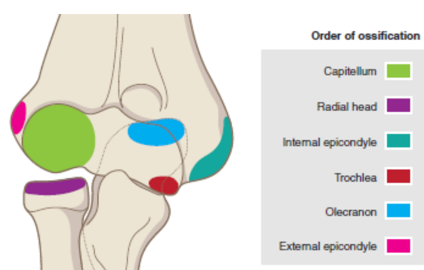
Fig. 15.29

Extrapolating for Interpretations

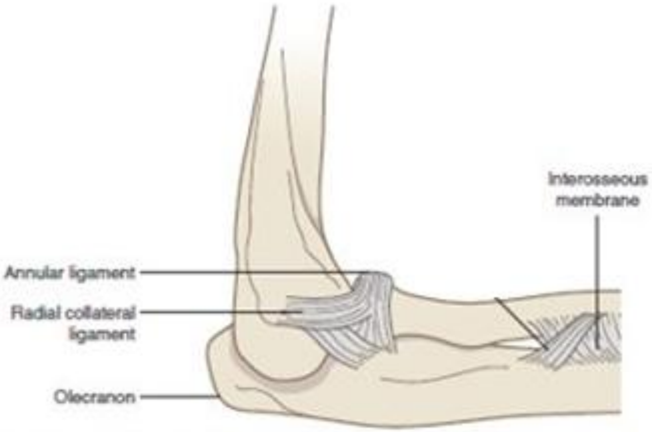


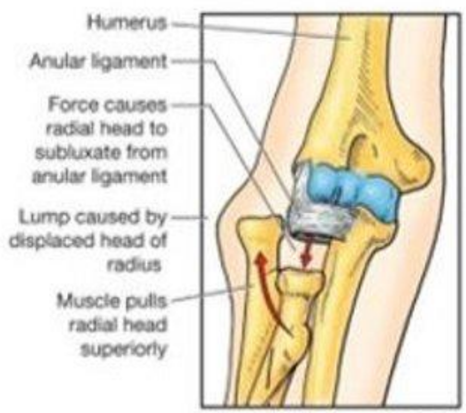
__Anterior humeral line__: a line drawn from the anterior surface of the **humerus** should intersect the middle third of the capitulum

__Radiocapitellar line__: a line drawn down the neck of the **radius** should intersect the capitulum

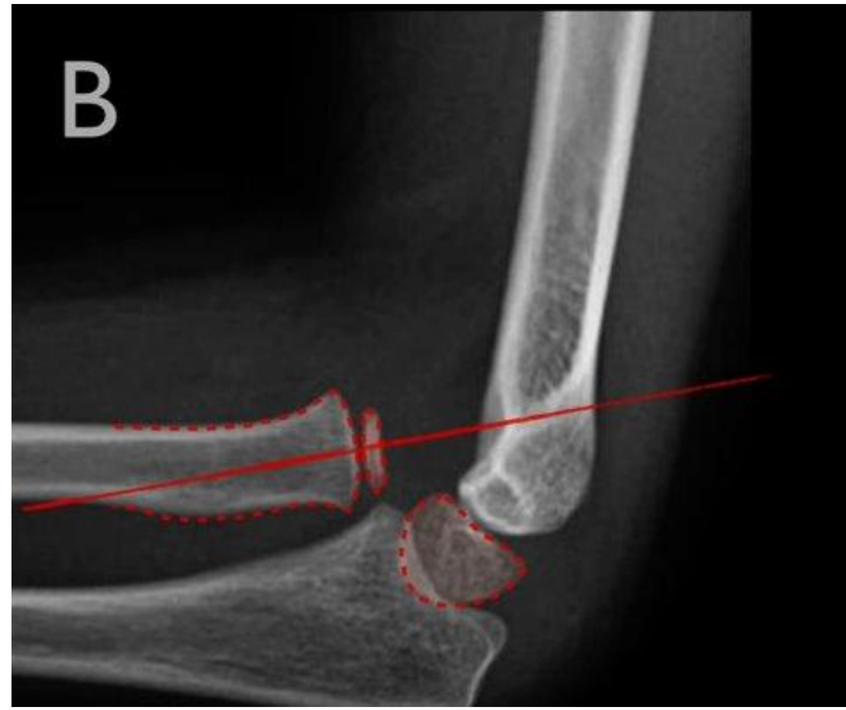


Clinical Correlate: “Nursemaid’s”/Pulled Elbow



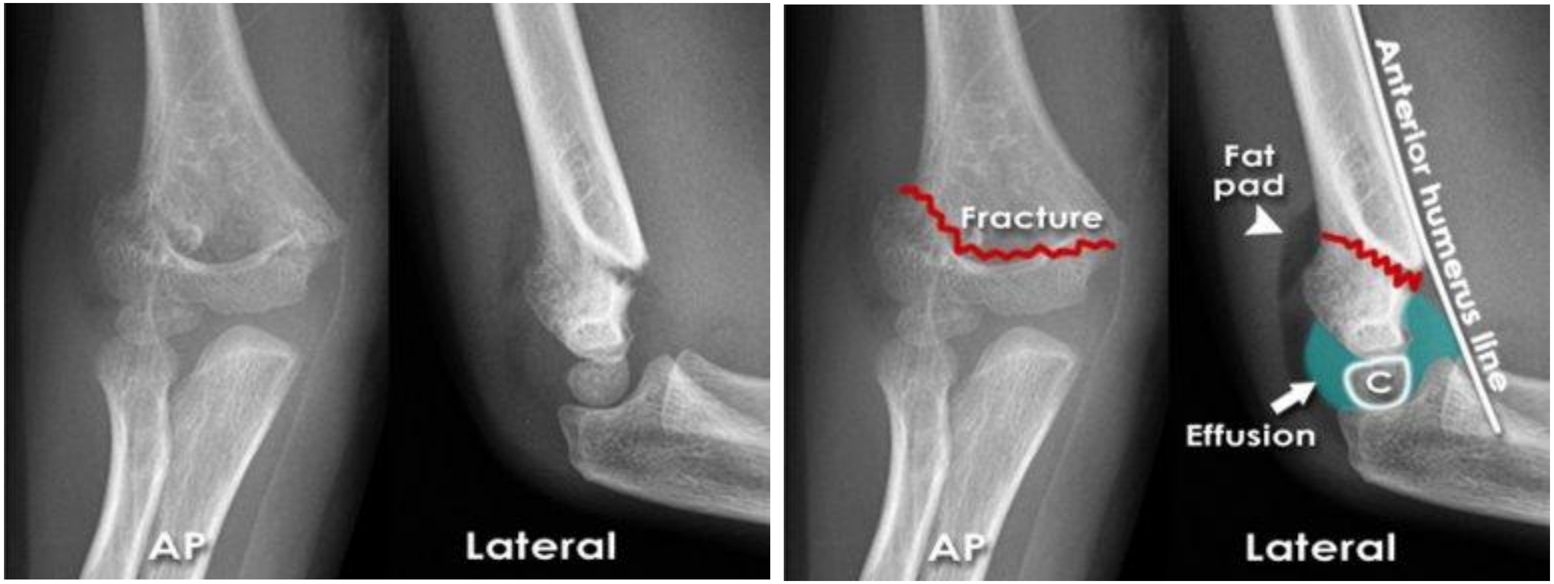


Subluxation and dislocation
Anterior view



Radiocapitellar line does not intersect the capitulum

Supracondylar fracture



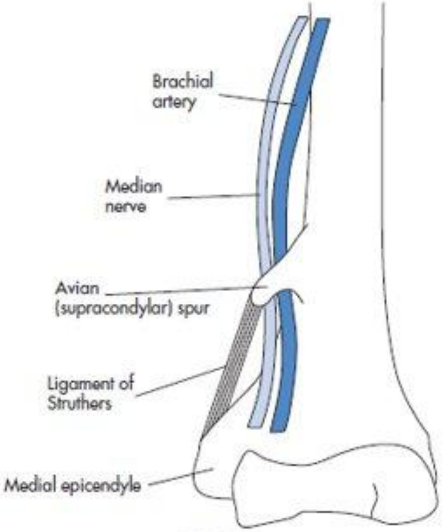
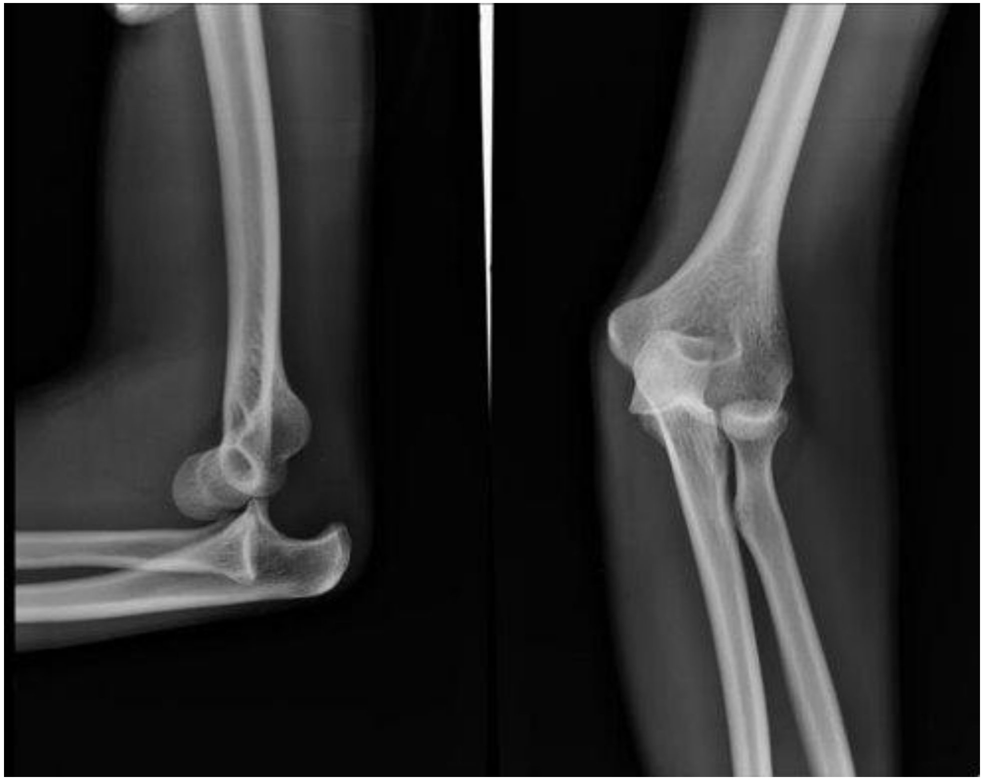


FIG. 5.55

Elbow dislocation



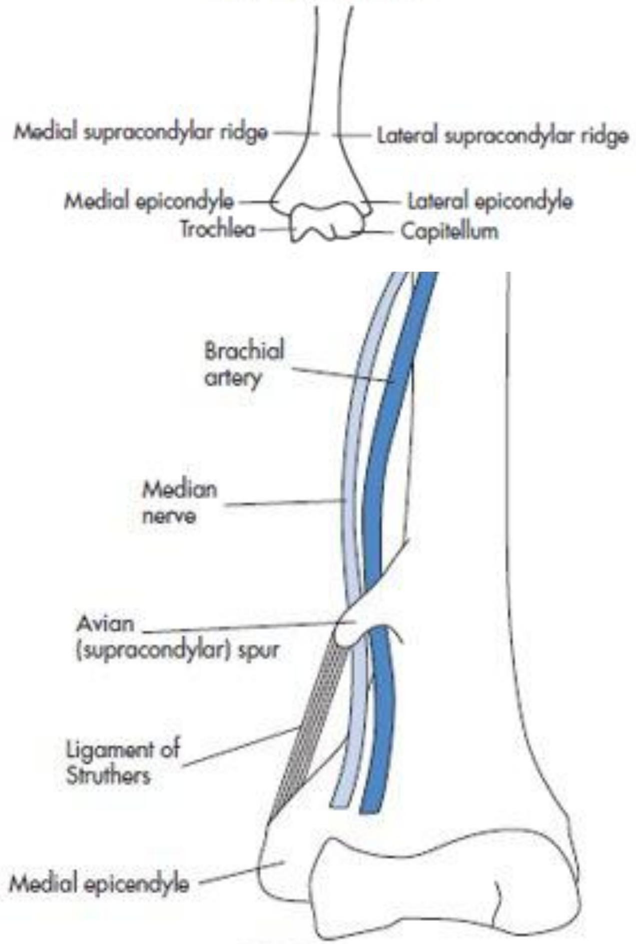
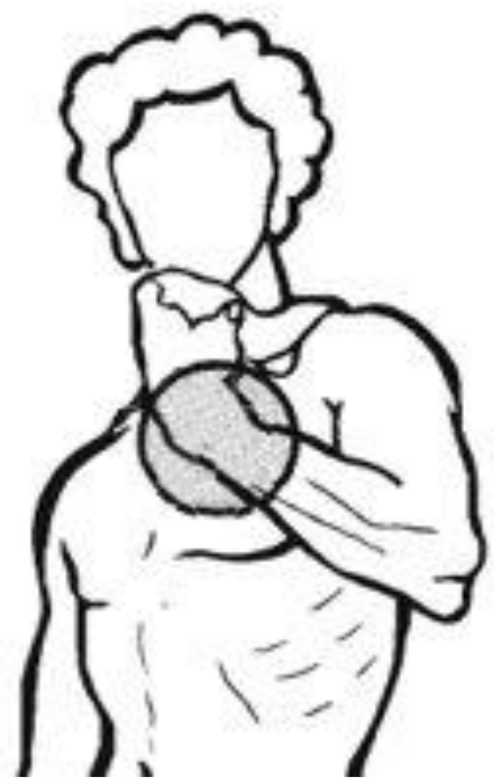


FIG. 5.55

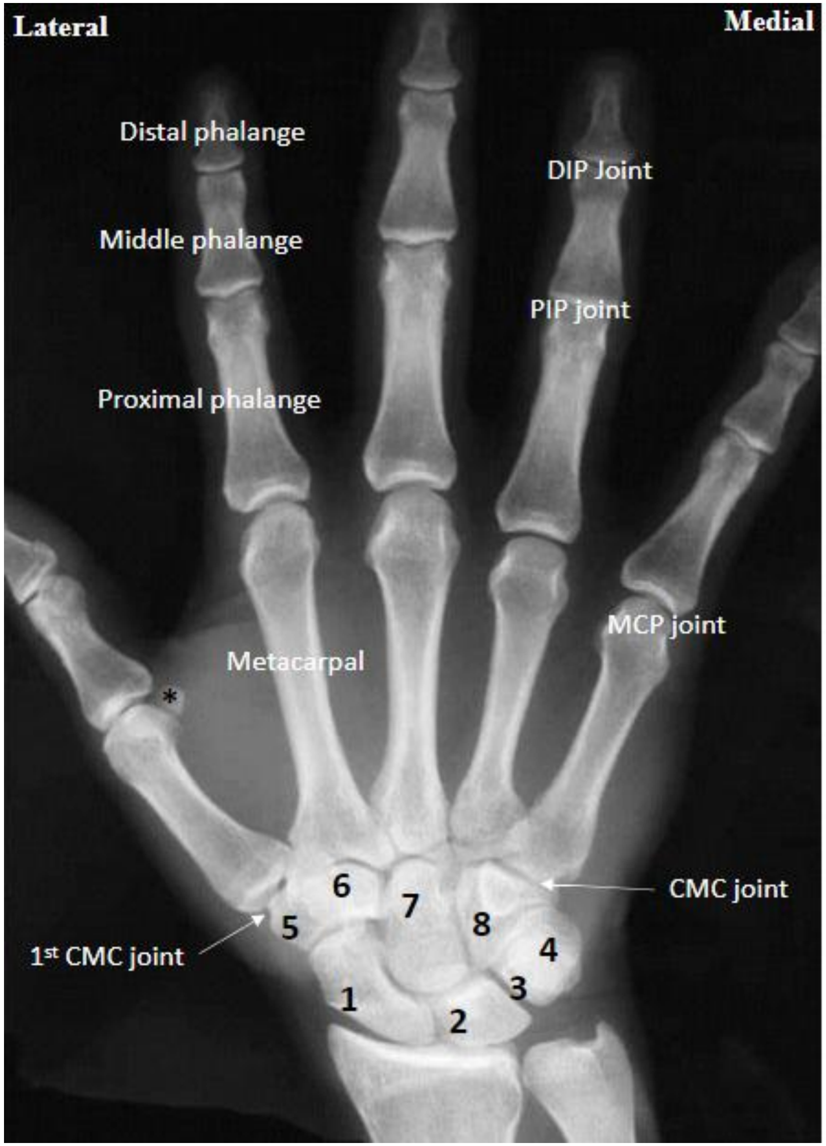
- Posterior dislocation of the elbow joint
- Possible damage to ulnar nerve, median nerve, brachial artery

Wrist



Wrist Imaging

Wrist - review



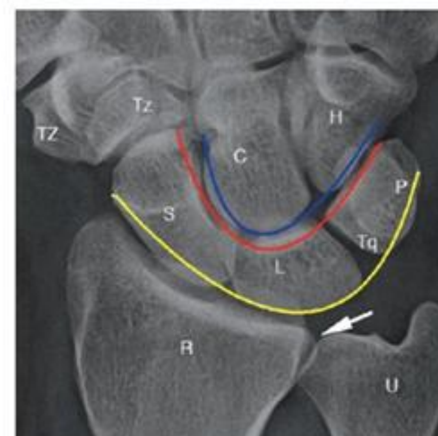


FIGURE 9.1

Clinical Correlate:
Fall Onto Outstretched Hand (FOOSH)

- *Colle's fracture*: fracture of distal radius with **posterior displacement** of the distal fragment
- *Smith's fracture*: fracture of the distal radius with **anterior displacement** of the distal fragment
- *Scaphoid fracture*: usually around the neck, dangerous possibility of avascular necrosis

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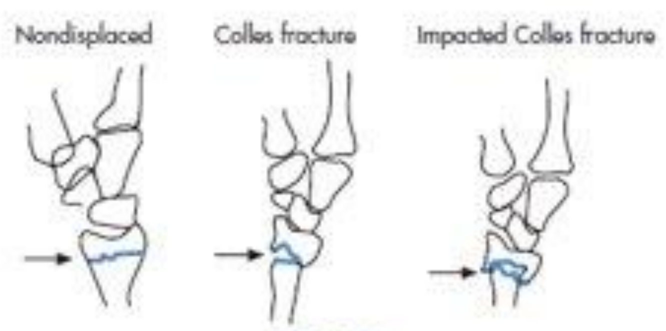
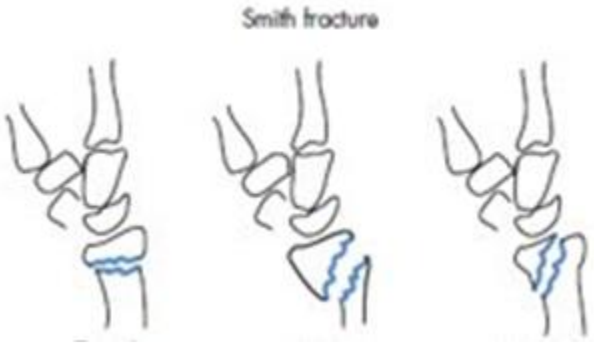
Clinical Correlate: Colles' Fracture

- **"Dinner fork" deformity** of the distal forearm
  - distal fracture fragment is displaced posteriorly
- May have injury to the median and/or ulnar nerve
  - Possible acute carpal tunnel syndrome
- Can compromise the radial or ulnar arteries and impair vascular supply to the hand



Clinical Correlate: Smith's fracture

- **"Garden spade" deformity** of the distal forearm
  - distal fracture fragment is displaced anteriorly
- Aka "reverse Colles"
- May have injury to the median and/or ulnar nerve
  - Possible acute carpal tunnel syndrome

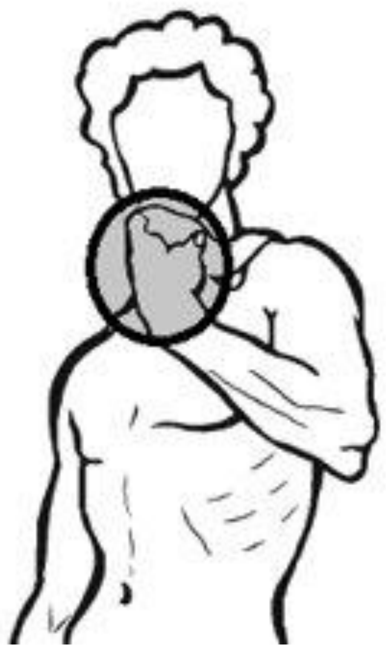


Clinical Correlate: Scaphoid Fracture

- Many scaphoid fractures remain occult on images taken
- **High-risk** of non-union, with or without avascular necrosis, of the proximal fracture component
    - In some patients, scaphoid receives its only blood supply from the radial artery
- Clinical suspicion may override radiological findings

FIGURE 4-17

Hand



Clinical Correlate: Boxer's fracture

- Impaction fracture of the neck of the fifth metacarpal
  - Sometimes can involve the fourth metacarpal
- Usually **comminuted**
- Most often due to a direct blow with a clenched fist against a solid surface

TABLE 24-3 HOW FRACTURES ARE DESCRIBED

Parameter	Terms Used
Number of fracture fragments	Simple or comminuted
Direction of fracture line	Transverse, oblique (diagonal), spiral
Relationship of one fragment to another	Displacement, angulation, shortening and rotation
Open to the atmosphere (outside)	Closed or open (compound)

FIGURE 2.7 •

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